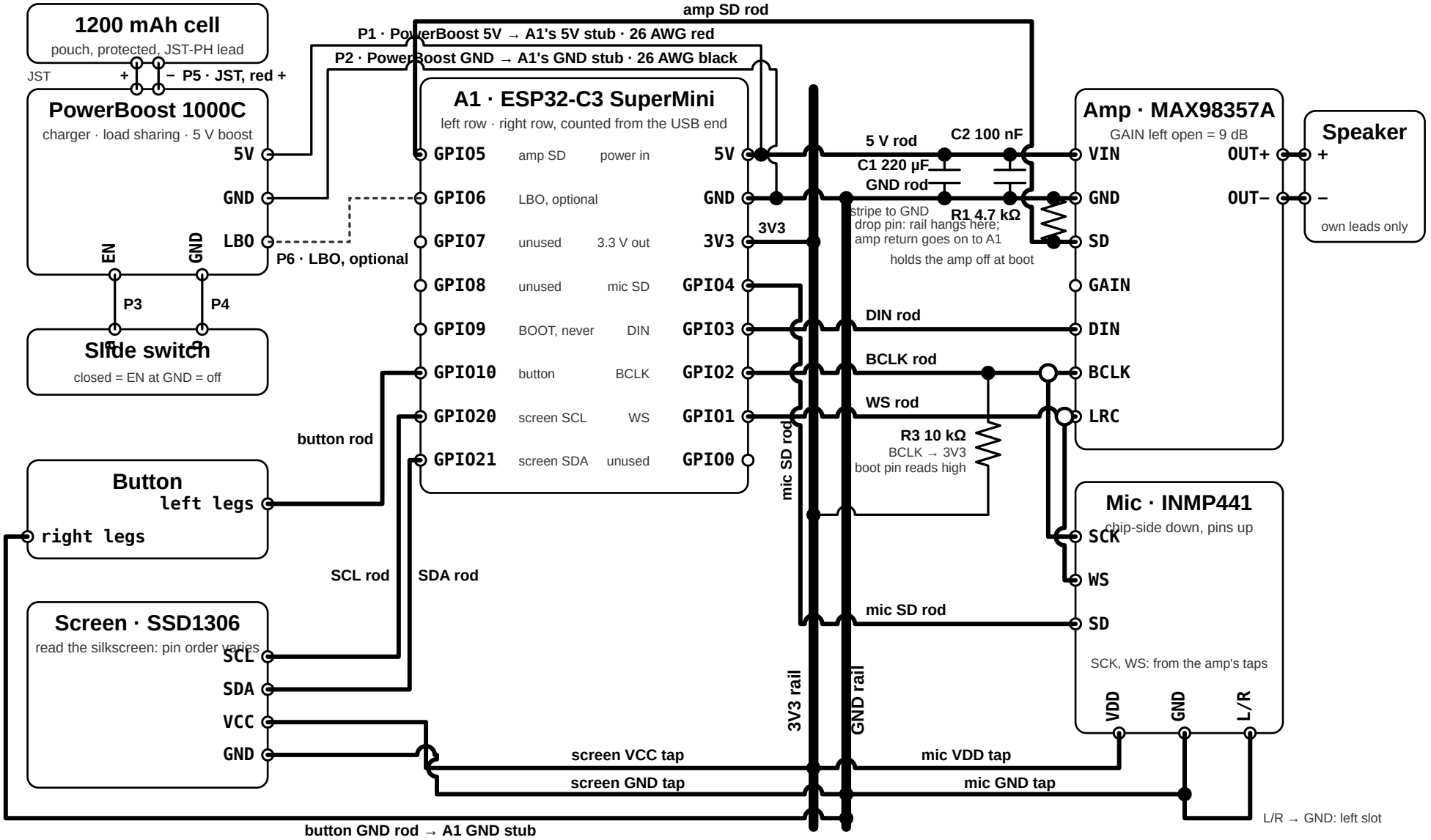
Pip, every connection on one sheet

A dot is a joint. A line that hops over another does not touch it. Thick = bare rod of the harness. Thin = insulated wire of the battery storey. Dotted = optional. Ring = tap where a rod carries on.



Never: OUT+ or OUT− to GND, a rail or the frame · anything but A1's 3V3 feeding the screen or the mic
A1's USB and the PowerBoost output at the same time · the frame touching any conductor · the 500 mAh cell

Build day: every connection, and why

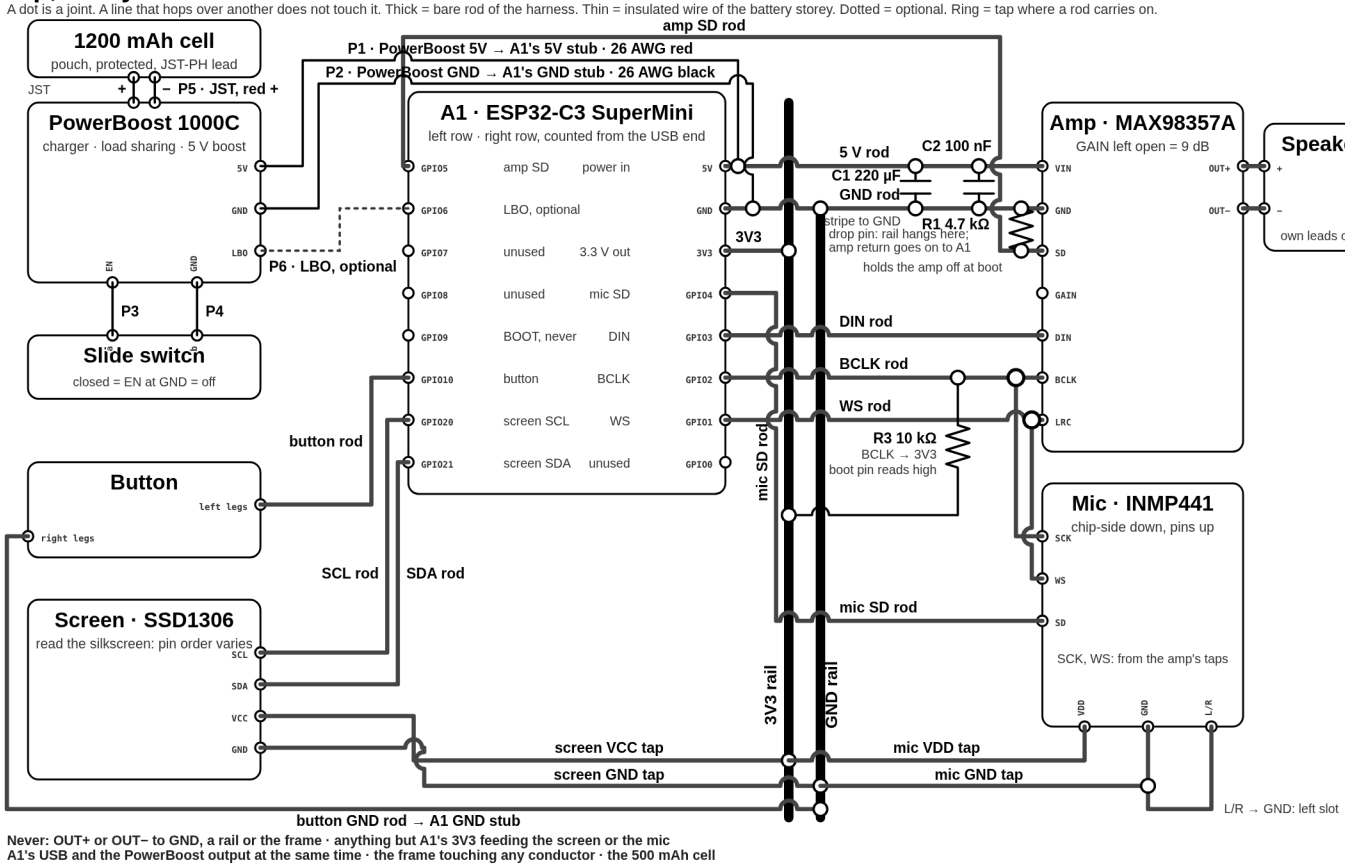
One sheet for the whole device. The brass harness first, on USB, on its jig card; then the PowerBoost, the switch and the cell underneath it. Every line says what joins what and the one reason it is done that way. Cut lengths, bends and per-rod drawings are on the step-by-step page; the battery details on the power page.

> **FOUR RULES FOR THE DAY**
>
> - **USB unplugged whenever the iron is on.** A powered board and a grounded iron tip are a short waiting to happen.
> - **OUT+ and OUT− go to the speaker only.** Both are switched outputs that sit at 5 V half the time; tied to ground, the amp dies in milliseconds.
> - **The frame carries nothing.** 2 mm of air or Kapton between any conductor and the brass; a grounded cage also detunes the antenna.
> - **One source at a time.** A1's 5V pin is its USB VBUS with no diode. PowerBoost on means A1's USB stays empty; flashing means PowerBoost off.

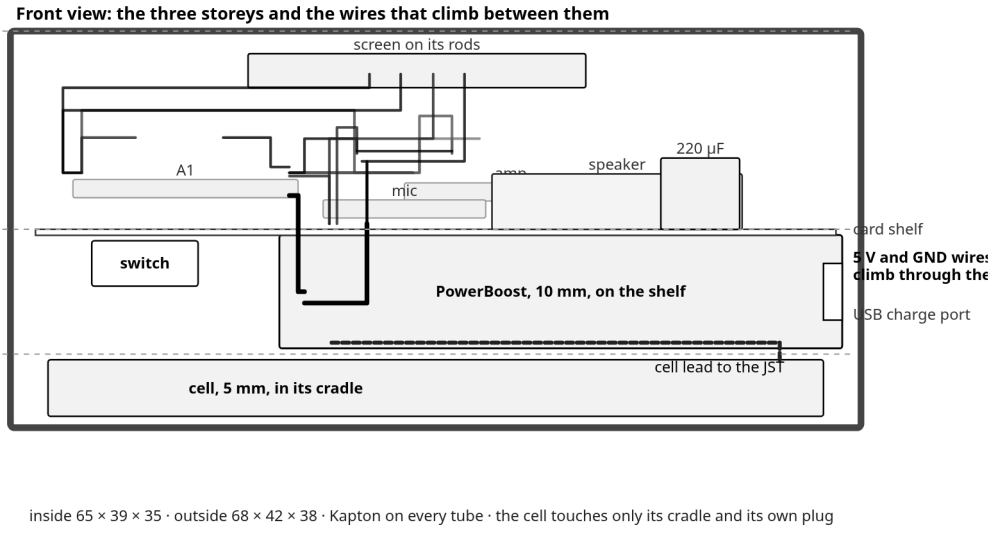
On the bench

Modules: A1 SuperMini, MAX98357A amplifier, INMP441 mic, SSD1306 screen, Same Sky CES-20134-088PM, tactile button. **Small parts:** 4.7 kΩ and 10 kΩ from the resistor kit, 220 µF from the electrolytic kit, 100 nF from the ceramic kit. **Conductors:** 1.0 mm brass rod or 18 ga nickel silver; 26 AWG and 30 AWG insulated wire for the battery storey. **Battery:** PowerBoost 1000C copy, 1200 mAh cell, slide switch. **Tools and consumables:** BOENFU flush cutters, WORKPRO pliers, BAISDY 400 grit, Chip Quik CQ4LF pen, MAIYUM 63/37 solder, JoTownCand solder wick, Pointool heat-shrink, ELEGOO Kapton tape, cardstock, KAIWEETS multimeter, NEIKO caliper, isopropyl, hot glue, heat gun, helping hands.

The two pictures to keep beside you



Every connection on one sheet. A1 in the middle with each GPIO named; thick lines are the bare rods of the harness, thin lines the insulated wires of the battery storey; a dot is a joint, a ring a tap where a rod carries on, a crossing without a dot does not touch. Printed landscape as page 1 of the PDF.



The whole device from the front, three storeys. Cell at the bottom in its cradle, the PowerBoost on a card shelf above it, the harness on top. Two insulated wires climb through the shelf to A1's 5V and GND stubs.

1 · The two rails

A rail is one bare rod that many parts share, so nine power and ground connections become nine short taps instead of nine rods back to A1. Every ground in the device is then the same piece of metal, which is what makes every signal read correctly: a 1 or a 0 is a voltage measured against ground, and two grounds that differ by a few tenths of a volt turn 1s into 0s.

RAIL	LIES	FED BY	WHO HOOKS ON	WHY
3V3 rail	x = 24, 21 mm, on the card	A1 3V3 stub, by the 3V3 tap	screen VCC, mic VDD, R3	A1's regulator makes 3.3 V; the mic dies above 3.6 V and the screen is a 3.3 V part
GND rail	x = 27, 22 mm, on the card	A1 GND stub, by the GND rod and a 3 mm drop pin	screen GND, mic GND, mic L/R, 220 µF –	the amp's bursty return takes the straight rod to A1; the quiet rail hangs off it, so mic and screen never share its noise

2 · Every A1 pin, where it goes, and why

A1 PIN	GOES TO	BY	WHY IT IS WIRED THAT WAY
`5V`	amp VIN	5 V rod, 10 mm, straight across the channel	only two pins ever see 5 V: this input and the amp. The amp wants 5 V because loudness is voltage: twice the swing of 3.3 V, four times the sound
`GND`	amp GND, then the GND rail	GND rod, 10 mm; a 3 mm drop pin under it onto the rail	the star point. Amp current returns on its own rod; the pin beneath makes the rail ground without routing that current along it
`3V3`	3V3 rail	3V3 tap, 7 mm, hooks down onto the rail	A1 feeds the rail once; nobody else runs back to A1 for power
`GPIO4`	mic SD	mic SD rod, 32 mm	sound in. The mic only drives this line during its slot; the firmware's internal pull-down holds it quiet the rest of the time (R4 dropped)
`GPIO3`	amp DIN	DIN rod, 10 mm	sound out. One listener, one rod
`GPIO2`	amp BCLK, then mic SCK	BCLK rod, 38 mm, one rod with two taps	the bit clock is the metronome; amp and mic must hear the same one. R3 10 kΩ to 3V3 sits on it because GPIO2 is a boot-strapping pin that must read high at power-on
`GPIO1`	amp LRC, then mic WS	WS rod, 33 mm, one rod with two taps	the word clock that says "left slot, right slot" 16,000 times a second; shared for the same reason
`GPIO5`	amp SD	amp SD rod, 43 mm	amplifier on/off. R1 4.7 kΩ to GND holds it off until the firmware is running, so the speaker cannot pop or sit at full gain on garbage
`GPIO10`	button, left legs	button rod, 19 mm; right legs to A1 GND by an 8 mm rod	pressing grounds the pin; A1 pulls it up internally. Plain high-or-low, no timing
`GPIO20`	screen SCL	SCL rod, 53 mm, up to the floating screen	I^2C clock. The pull-ups I^2C needs are already on the screen module, so the harness has none
`GPIO21`	screen SDA	SDA rod, 53 mm	I^2C data, same bus

Left as stubs: GPIO0, 6, 7, 8, 9. GPIO6 is reserved for the PowerBoost's low-battery flag; GPIO9 is the BOOT button and never an input.

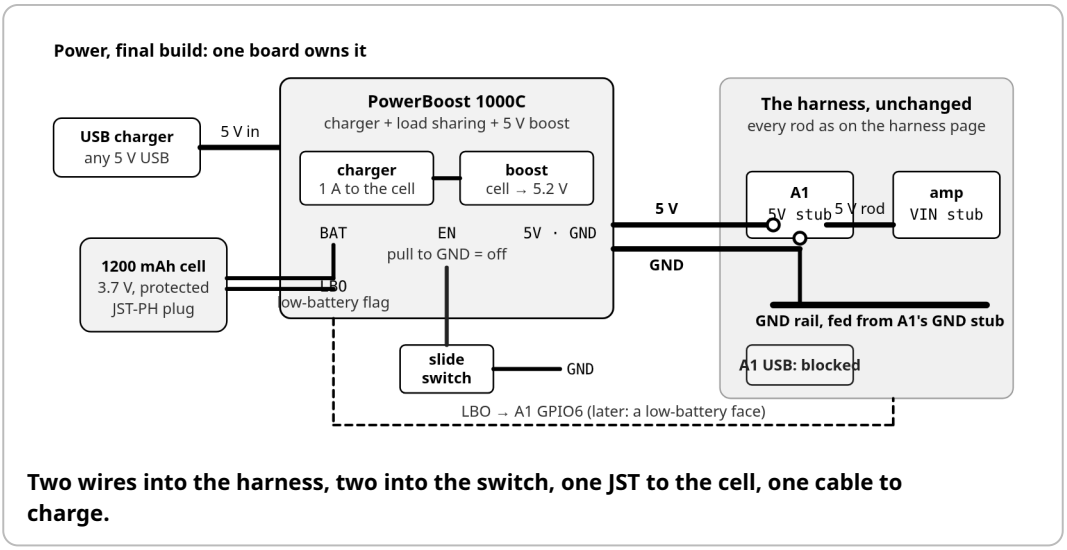
3 · Taps and the small parts

PART PIN	TO	BY	WHY
screen GND · VCC	GND rail front · 3V3 rail front	two taps, 23 and 22 mm, dropping from the floating screen	the screen is held up by its four rods; power comes from the rails nearest it
mic GND · VDD	GND rail back · 3V3 rail back	two taps, 11 and 20 mm	the mic lies chip-side down at the back; its SD, VDD, GND row faces the channel
mic L/R	mic GND stub	8 mm link	grounding L/R selects the left I^2S slot, the one the firmware listens to
amp GND	A1 GND stub	the GND rod	see the star point above
R1 4.7 kΩ	amp SD ↔ amp GND	a U lying over the amp board	keeps the amp silent through the undefined moment between power-on and firmware
R3 10 kΩ	BCLK rod ↔ 3V3 rail	in the channel, 8.5 mm up	boot-strap pull-up for GPIO2. One meter check retires it: A1 unpowered, ~10 kΩ between GPIO2 and 3V3 means the board has its own
C1 220 µF	5 V net ↔ GND rod, stripe to GND	lying in the right strip, leads sleeved	a local reservoir for bass notes and Wi-Fi bursts, so the 5 V does not sag and reset A1
C2 100 nF	amp VIN ↔ amp GND	flat on the amp board at the pins	answers the fast spikes the 220 µF is too slow for; it must sit at the pins to work
speaker	OUT+ and OUT− only	its own two leads, twisted	a bridge output; neither side is ground, ever

Dropped on 2026-10-14: R2 (GAIN stays open, 9 dB; add 100 kΩ GAIN-to-GND later for 15 dB if Pip is too quiet) and R4 (the firmware enables GPIO4's pull-down).

4 · The battery storey

A lithium cell gives 3.7 V falling to 3.0 V. Fed straight to A1 the amp is a whisper and A1's regulator browns out with half the cell still full. The PowerBoost does three jobs the harness cannot: boost the cell to a steady 5.2 V so the harness sees exactly what USB gave it, charge the cell correctly (1 A until 4.2 V, then hold 4.2 V until the current tapers), and share the load so the device can run while charging. The harness does not change; its two power entry points just get a new supplier.



Two wires into the harness, two into the switch, one JST to the cell, one cable to charge.

#	FROM	TO	WITH	WHY
P1	PowerBoost `5V`	A1's 5V stub, through the shelf hole	26 AWG red, ≈ 60 mm	that stub is where USB power used to enter; every rod downstream sees the same 5 V it always did, and the 220 µF stays where it is
P2	PowerBoost `GND`	A1's GND stub, through the shelf hole	26 AWG black, ≈ 55 mm	one ground meeting point. The rail stays a dead end, so no loop for the amp's current to run through the signal rods
P3	PowerBoost `EN`	slide switch, one side	30 AWG, ≈ 40 mm	EN is a logic input drawing microamps, so a tiny switch controls the whole load without carrying its 0.7 A or dropping volts where volts are scarce
P4	switch, other side	PowerBoost `GND`	30 AWG, ≈ 40 mm	closed = EN grounded = output off. The cell stays on the charger even when "off", so it charges either way
P5	1200 mAh cell JST	PowerBoost JST socket	the cell's own lead	the plug is the only connection the cell will ever have: factory-terminated, keyed, behind its own protection board. Meter + and − against the silkscreen before the first plug-in
P6	PowerBoost `LBO`	A1 GPIO6	30 AWG, optional	open-drain flag that goes low at 3.2 V; a free GPIO reads it so the face can warn before the cell cuts out
—	A1 USB-C	nothing; Kapton over it	—	A1's 5V stub is the USB VBUS node with no diode; a cable there would tie two supplies together

Why the 1200 mAh and not the 500. The device peaks near 0.5 A at 5 V; through the boost that is about 0.75 A from the cell, 1.5C for a 500 mAh cell and over its rating, 0.6C for the 1200 and comfortable. Charging at 1 A is normal for the 1200, abusive for the 500. **Why no solder ever touches the cell.** It is a foil envelope; heat the tab and the separator inside can melt. It goes in last and comes out first.

5 · The order of the day

1. **Stubs.** Clip A1's and the amp's pins flush under their header strips; the short ends on top are the stubs. Screen and mic pins to 2 mm, button legs to 1 mm. Sand every stub bright. *Why:* a rod lies against a short stub; a long pin would push it off the card.
2. **Jig card.** Print the floor template at Actual size, glue to cardstock. Kapton on the USB socket roof (two layers) and a strip on the amp beside its stubs. Tape the modules on their outlines the way up the step page says; shim the amp's pin-free edge to 2.5

mm. *Why:* everything is positioned on the card and soldered in place; the card comes off at the end.

3. **Rails.** 3V3 at x = 24, GND at x = 27, taped. Meter between them: silent.
4. **Six channel rods.** 5 V rod, GND rod, GND drop pin, 3V3 tap, DIN rod, then the BCLK and WS rods with their amp taps. Flux, 1–2 s of iron per joint, 320 °C. *Why this order:* low rods first; everything later crosses above them.
5. **Four fly-overs.** Mic SD, amp SD, button rod, and the BCLK and WS continuations to the mic, up to 10.5 mm. Heat-shrink wherever one crosses another under 1 mm.
6. **Screen** on its four rods (SCL, SDA, VCC tap, GND tap), floating 17.5 mm up. **Button** on the socket's Kapton. **Mic** chip-side down, its three taps and the L/R link.
7. **R1, R3, C1, C2, speaker.** Stripe of the 220 µF to GND. Speaker leads to OUT+ and OUT− only.
8. **Power up on USB.** Meter sweep first: GND rail to 3V3 rail, to the 5 V rod, to OUT+, to OUT−, all silent. USB in; eyes open, `got ip`, `hello accepted` in the server log. Hold the button, ask Pip its name. Then the amplifier build at cap 45 and Pip answers out loud. Unplug, wait ten seconds, plug in, ask once more.
9. **Off the card.** Cut the tape and lift the harness as one rigid piece. Kapton inside the frame. Meter frame to every rail and to OUT±: silent.
10. **Bench-test the PowerBoost alone** before it meets the harness: cell in (red to +), switch on EN, meter 5.1–5.3 V at its output, then one charge watching the LEDs (yellow charging, green full). The eight steps are on the real-test page.
11. **Assemble the storeys.** Shelf with its two wire holes; PowerBoost on it, USB port to the wall; P1–P4 soldered at the PowerBoost end first and led up through the holes; harness onto the shelf; P1 to A1's 5V stub, P2 to A1's GND stub; Kapton over A1's USB; cradle and cell at the bottom, JST in last.
12. **Final checks.** Meter frame to everything: silent. Switch on: the same tests as step 8, now with no cable. Charge once through the PowerBoost's port with the device on.

> **NEVER**
>
> - OUT+ or OUT− to GND, a rail, the frame, or a scope ground clip.
> - Anything but A1's 3V3 pin feeding the screen or the mic.
> - A1's USB and the PowerBoost output at the same time.
> - Solder, brass, or an iron within 10 cm of the cell. The 500 mAh cell at all.
> - Harris acid flux on this bench: it is for the empty frame only, washed and neutralised before any module arrives.

Sources. Connections and reasons from the net list, Why the Wiring, and the final-build power page, all generated from `harness_model.py` and the CLAUDE.md pin contract; PowerBoost functions from Adafruit's product page; cell ratings from `docs/INVENTORY.md`. Currents and runtime are estimates, not measurements. The harness and the battery storey have not been built.